

IBM Business Automation Manager Open Editions Demos and Labs 2026

Expose your DMN project with MCP Server

V 2.0 (for IBM BAMOE 9.5.2)

Raul Mariano
raul.mariano@ibm.com

NOTICES

This information was developed for products and services offered in the USA.

IBM may not offer the products, services, or features discussed in this document in other countries. Consult your local IBM representative for information on the products and services currently available in your area. Any reference to an IBM product, program, or service is not intended to state or imply that only that IBM product, program, or service may be used. Any functionally equivalent product, program, or service that does not infringe any IBM intellectual property right may be used instead. However, it is the user's responsibility to evaluate and verify the operation of any non-IBM product, program, or service.

IBM may have patents or pending patent applications covering subject matter described in this document. The furnishing of this document does not grant you any license to these patents. You can send license inquiries, in writing, to:

IBM Director of Licensing
IBM Corporation
North Castle Drive, MD-NC119
Armonk, NY 10504-1785
United States of America

The following paragraph does not apply to the United Kingdom or any other country where such provisions are inconsistent with local law: INTERNATIONAL BUSINESS MACHINES CORPORATION PROVIDES THIS PUBLICATION "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Some states do not allow disclaimer of express or implied warranties in certain transactions, therefore, this statement may not apply to you.

This information could include technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in new editions of the publication. IBM may make improvements and/or changes in the product(s) and/or the program(s) described in this publication at any time without notice.

Any references in this information to non-IBM websites are provided for convenience only and do not in any manner serve as an endorsement of those websites. The materials at those websites are not part of the materials for this IBM product and use of those websites is at your own risk.

IBM may use or distribute any of the information you supply in any way it believes appropriate without incurring any obligation to you.

Information concerning non-IBM products was obtained from the suppliers of those products, their published announcements or other publicly available sources. IBM has not tested those products and cannot confirm the accuracy of performance, compatibility or any other claims related to non-IBM products. Questions on the capabilities of non-IBM products should be addressed to the suppliers of those products.

This information contains examples of data and reports used in daily business operations. To illustrate them as completely as possible, the examples include the names of individuals, companies, brands, and products. All of these names are fictitious and any similarity to the names and addresses used by an actual business enterprise is entirely coincidental.

TRADEMARKS

IBM, the IBM logo, and ibm.com are trademarks or registered trademarks of International Business Machines Corp., registered in many jurisdictions worldwide. Other product and service names might be trademarks of IBM or other companies. A current list of IBM trademarks is available on the web at "Copyright and trademark information" at www.ibm.com/legal/copytrade.shtml.

Adobe, the Adobe logo, PostScript, and the PostScript logo are either registered trademarks or trademarks of Adobe Systems Incorporated in the United States, and/or other countries.

Cell Broadband Engine is a trademark of Sony Computer Entertainment, Inc. in the United States, other countries, or both and is used under license therefrom.

Intel, Intel logo, Intel Inside, Intel Inside logo, Intel Centrino, Intel Centrino logo, Celeron, Intel Xeon, Intel SpeedStep, Itanium, and Pentium are trademarks or registered trademarks of Intel Corporation or its subsidiaries in the United States and other countries.

IT Infrastructure Library is a Registered Trade Mark of AXELOS Limited.

ITIL is a Registered Trade Mark of AXELOS Limited.

Java and all Java-based trademarks and logos are trademarks or registered trademarks of Oracle and/or its affiliates.

Linear Tape-Open, LTO, the LTO Logo, Ultrium, and the Ultrium logo are trademarks of HP, IBM Corp. and Quantum in the U.S. and other countries.

Linux is a registered trademark of Linus Torvalds in the United States, other countries, or both.

Microsoft, Windows, Windows NT, and the Windows logo are trademarks of Microsoft Corporation in the United States, other countries, or both.

UNIX is a registered trademark of The Open Group in the United States and other countries.

© Copyright International Business Machines Corporation 2024.

This document may not be reproduced in whole or in part without the prior written permission of IBM.

US Government Users Restricted Rights - Use, duplication or disclosure restricted by GSA ADP Schedule Contract with IBM Corp.

Table of Contents

1	Introduction.....	4
1.1	IBM Business Automation Manager Open Editions.....	4
2	Lab Setup Instructions	5
2.1	Access the environment.....	5
2.1.1	About the Environment:	5
3	Lab Objective	6
4	Exercise 1: Deploy the BAMOE MCP Server on Podman	7
5	Exercise 2: Create and test a Langflow agent integrated with your BAMOE MCP Server ...	11
6	Consult Documentation and Communities.....	20

1 Introduction

These hands-on lab exercises are designed to guide you through the essential aspects of process automation using BAMOE Development Tools for VS Code. Whether you're a developer or an architect, these labs will provide the skills needed to use BAMOE Canvas and VS Code to effectively create workflow automation and decision-making designs in modern, cloud-native business automation solutions.

Notes:

- *Includes **four** exercises. We recommend completing them sequentially.*
- **Duration:** *Approximately 1 hour.*
- **Target audience:** *Anyone who wants to learn how to use IBM Business Automation Manager Open Editions.*

1.1 IBM Business Automation Manager Open Editions

IBM Business Automation Manager Open Editions (IBM BAMOE) is a powerful open-source solution that serves as a foundation platform for tailoring long-lasting business automation solutions for the hybrid cloud.

With a developer-centric approach, this comprehensive and flexible platform makes it easy for teams to collaborate through Open Standards and efficient development tools suited for different personas.

Each automation solution can be shaped to perfectly address each scenario: business applications are flexible and can effortlessly integrate with external systems of your existing architecture.

Designed for the hybrid cloud, IBM Business Automation Manager Open Editions, accelerates the application modernization and cloud adoption journeys, as the lightweight design tools, business applications and other product components can be containerized and deployed with popular technologies such as Kubernetes and OpenShift.

For more information, see IBM documentation and other useful links:

- [IBM Business Automation Manager Open Editions Documentation](#)
- [Open Editions Community](#)

2 Lab Setup Instructions

2.1 Access the environment

You received this email with instructions on how to access the environment using your IBMid. Otherwise, click [here](#) to reserve it.

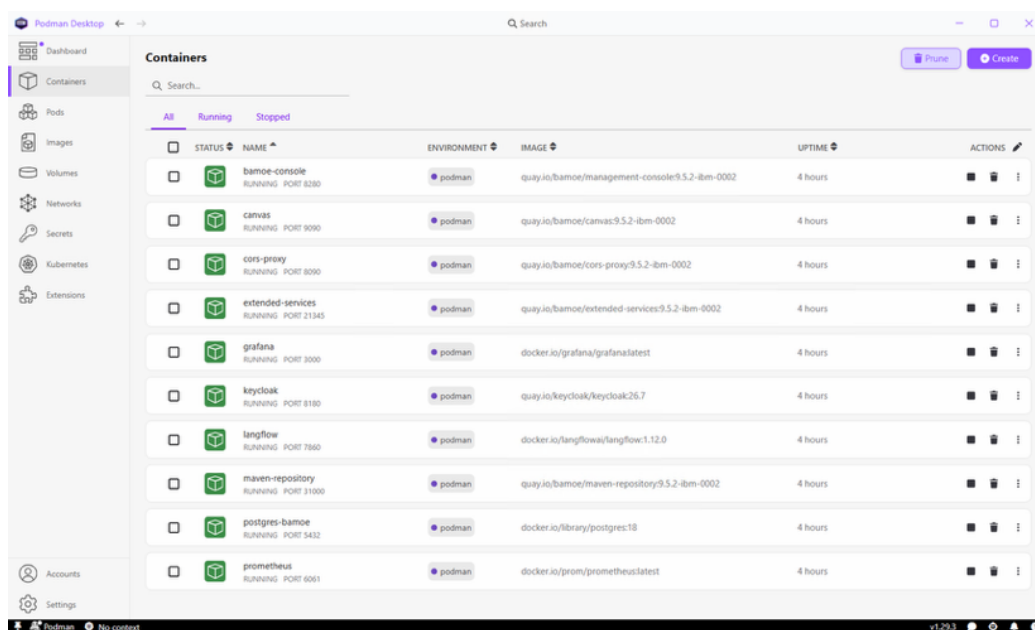
If necessary, this is the Windows credential:

```
User: techzone  
Password: IBMDemOs!
```

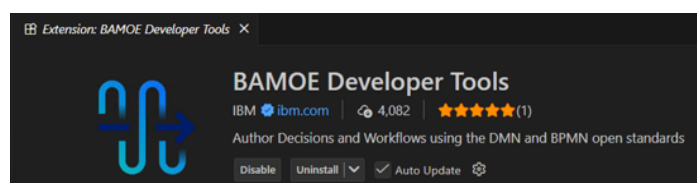
2.1.1 About the Environment:

We built this environment so that you can practically use IBM Business Automation Manager Open Editions, so all applications are installed locally. To perform this lab, it will not be necessary to use any external services. Basically, we have this architecture:

- [Development](#) and [Runtime](#) applications are running in containers on Podman.



- For deployment and testing of your applications, VS Code with the [BAMOE Developer Tools](#) extension will be used.



3 Lab Objective

See in practice how to use the BAMOE MCP Server to enable your agents to interact with your BAMOE applications. This feature is still in [Tech Preview](#), but it is already possible to have a good experience connecting Agents to DMNs. Consult the documentation and follow upcoming updates to stay informed about the latest developments.

This lab was divided into 2 main exercises, see below:

- Exercise 1 - Deploy the BAMOE MCP Server on Podman.
- Exercise 2 - Create and test a Langflow agent integrated with your BAMOE MCP Server.

To complete this lab, we assume you have already had some contact with IBM BAMOE; therefore, some navigation or explanations that we consider "basic" will not be covered in detail.

Next, the first exercise begins. Happy studying!

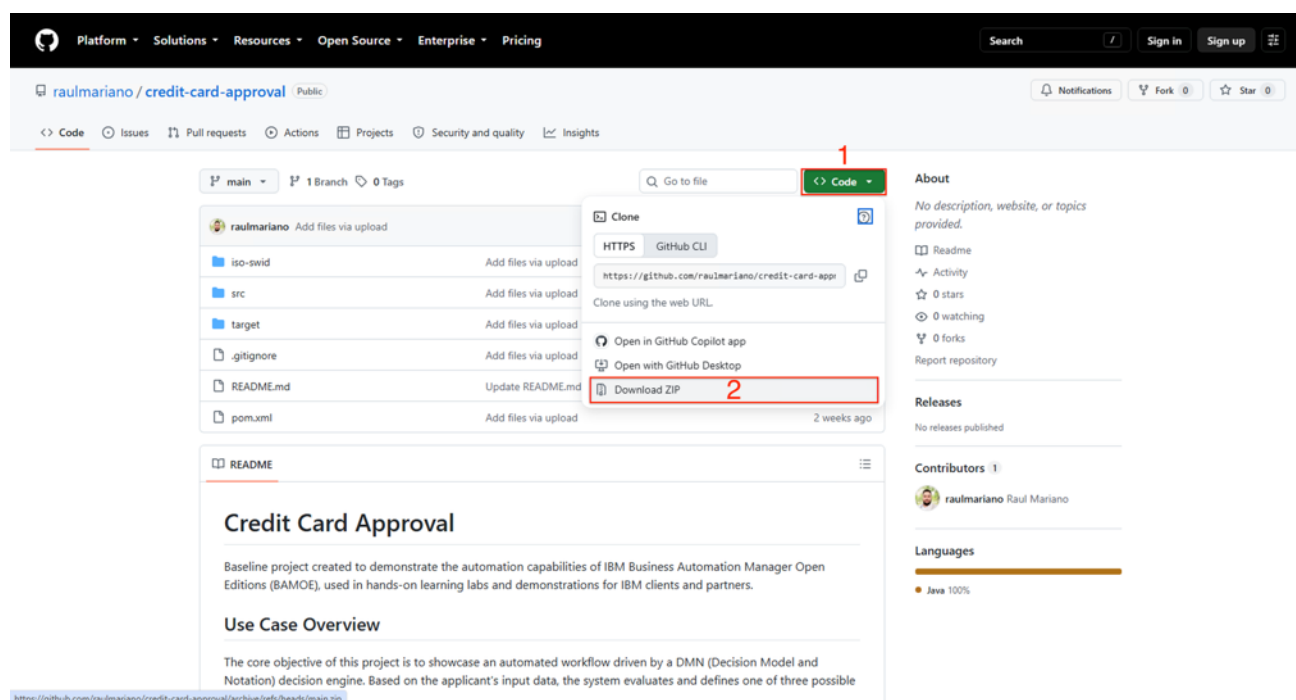
4 Exercise 1: Deploy the BAMOE MCP Server on Podman

We will use the same use case as in other labs, so you can download the Credit Card Approval project from GitHub.

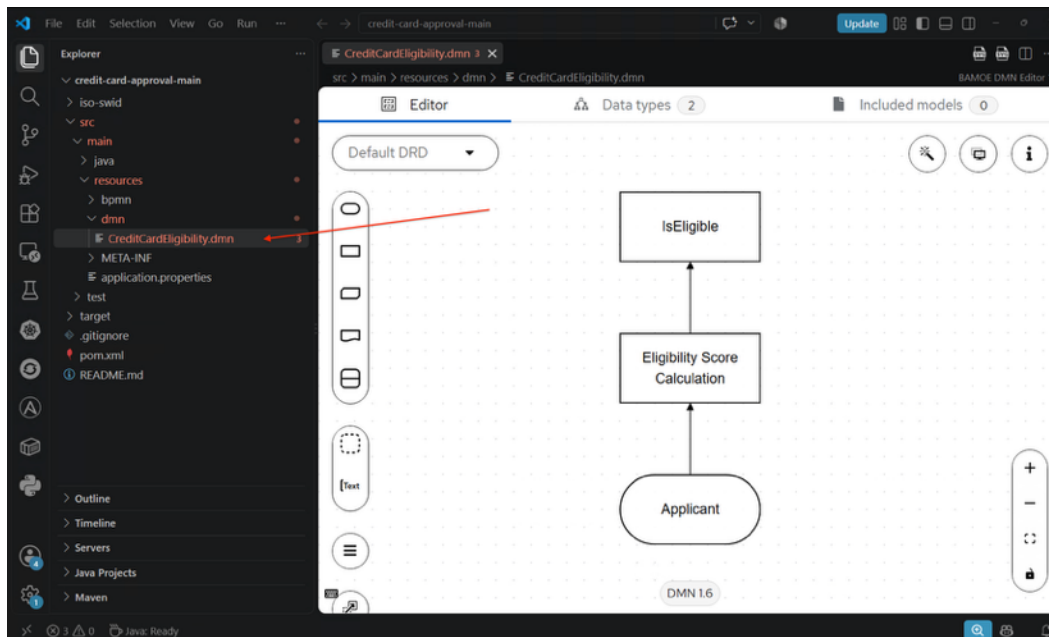
- The first step will be to download the project from GitHub. You can access the link and download the **.zip** file.

<https://github.com/raulmariano/credit-card-approval>

- Click on **"Code"** then on **"Download ZIP"**



- Save the **.zip** file and extract everything to a directory of your choice.
- Open the project in VS Code. If you are not yet familiar with this use case, explore the **"CreditCardEligibility.dmn"** DMN file in particular.



Since we are using a local environment, we will launch an MCP Server without security. Consult the [documentation](#) to implement a secure deployment. Therefore, we do not need to perform any additional configuration in the project.

What do you need to know?

- In the “*application.properties*” file, look for the `quarkus.smallrye-openapi.path` property, this is the OpenAPI path required to start the MCP Server. By default, this OpenAPI endpoint will be created for development execution (which is the scenario here).

```

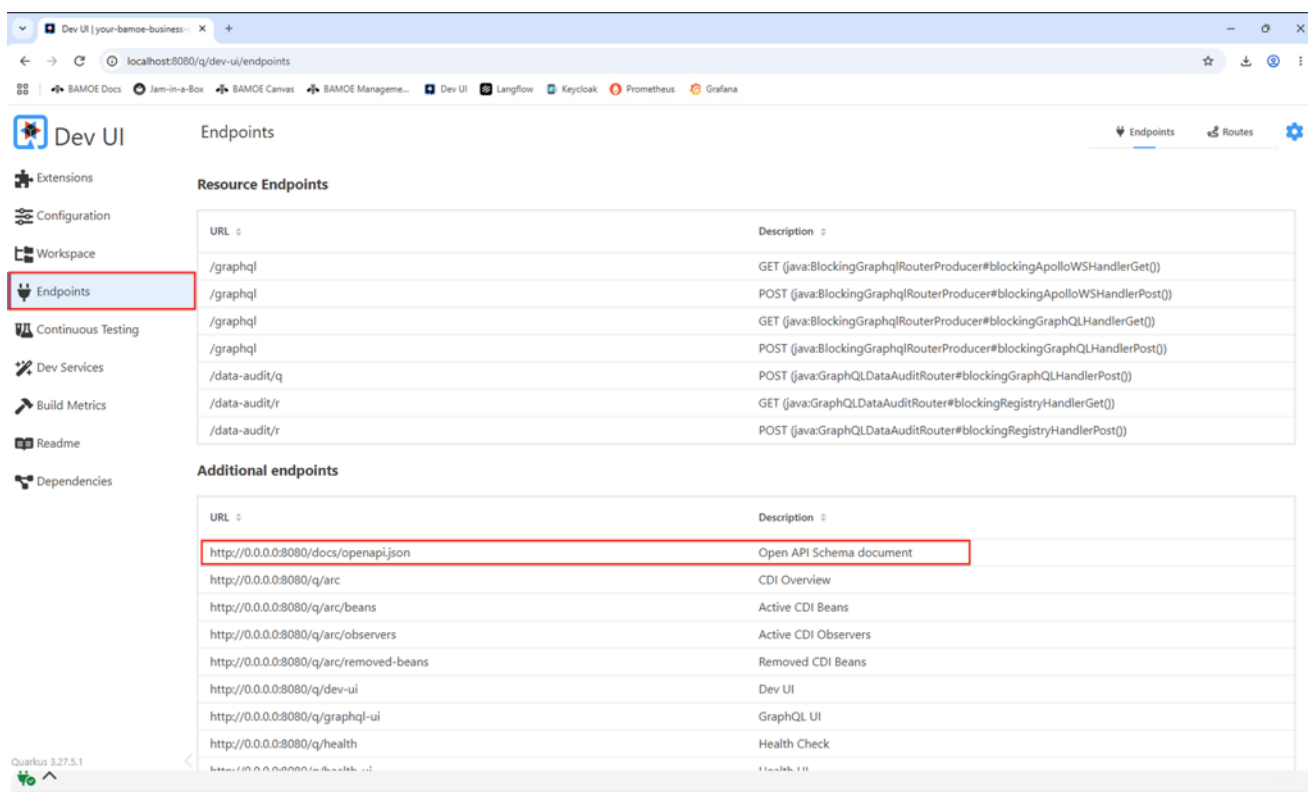
182
183 #####
184 # BAMOE AI Agent Task #
185 #####
186
187 # Langflow
188 # bamoe.workflow.ai-agent-task.provider.langflow.base-url=<LANGFLOW_BASE_URL>
189 # bamoe.workflow.ai-agent-task.provider.langflow.api-key=<API_KEY>
190 # bamoe.workflow.ai-agent-task.provider.langflow.log-requests=<true/false>
191 # bamoe.workflow.ai-agent-task.provider.langflow.log-responses=<true/false>
192
193 #####
194 # Misc. :: Dev #
195 #####
196
197
198 %dev.quarkus.smallrye-openapi.path=${quarkus.http.root-path}/docs/openapi.json //docs/openapi.js
199 %dev.quarkus.http.test-port=0
200 %dev.quarkus.dev-ui.cors.enabled=false
201 %dev.quarkus.swagger-ui.always-include=true
202 %dev.quarkus.kogito.data-index.graphql.ui.always-include=true
203 %dev.bamoe.devui.users.admin.groups=admin
204 %dev.bamoe.devui.users.jdoe.groups=jdoe
205
  
```

- Open a new terminal and let's run the project, use the following command:

```
mvn clean quarkus:dev -Pdevelopment "-Dquarkus.http.host=0.0.0.0"
```

```
PS C:\techjam\credit-card-approval-main\credit-card-approval-main> mvn clean quarkus:dev -Pdevelopment "-Dquarkus.http.host=0.0.0.0"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.acme:your-bamoe-business-service >-----
[INFO] Building your-bamoe-business-service 1.0.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
```

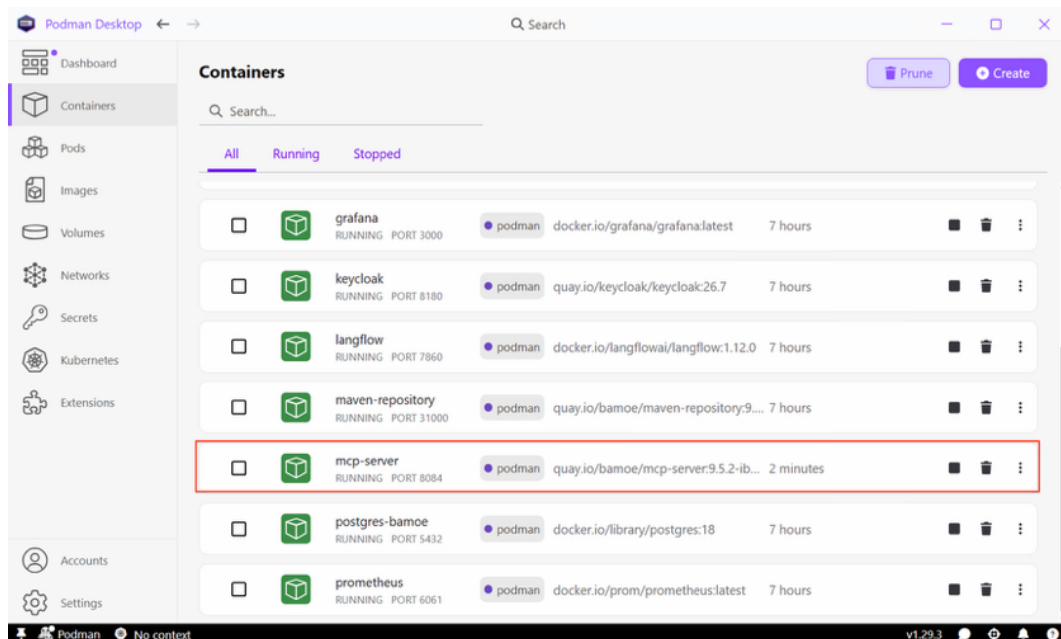
- In the Chrome browser, open the dev-ui and let's check the endpoints.
- Note that this path is available.



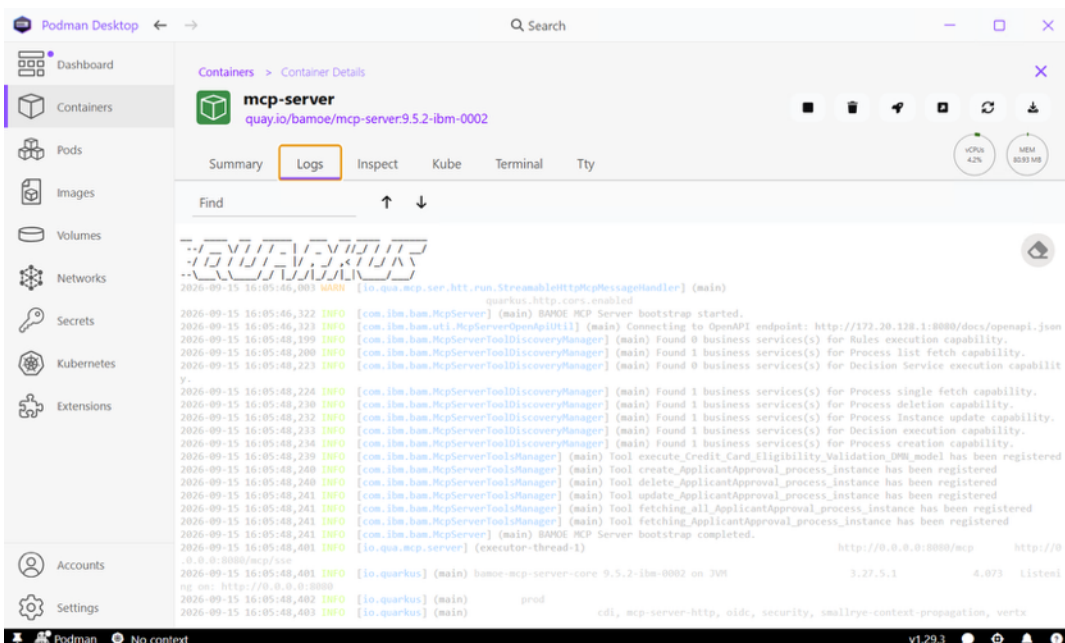
- Great, now that we have the project running, let's create the MCP Server. The MCP Server will be configured and deployed in a container on Podman, via the command line, we will pull an image and run it.
- In the windows terminal, run the command to start an MCP Server Container:

```
podman run -d -it --rm -p 8084:8080 --name mcp-server --network bamoe-net --add-host host.containers.internal:host-gateway -e MCP_SERVER_OPENAPI_URLS=http://172.20.128.1:8080/docs/openapi.json quay.io/bamoe/mcp-server:9.5.2-ibm-0002
```

- After running it, you can open Podman Desktop and check that the MCP Server is now available on port 8084.



- Click on the container to check for any errors during deployment.



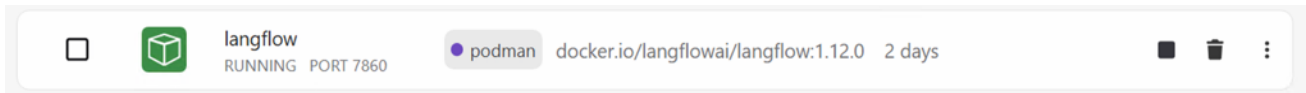
Okay, now any MCP client can consume the BAMOE MCP server, for example, Goose, 5ire, LM Studio, and Claude Desktop.

In the next exercise, we will use Langflow to test an interaction between a Chat Agent and the BAMOE MCP Server.

5 Exercise 2: Create and test a Langflow agent integrated with your BAMOE MCP Server

Now we're going to create a Langflow agent to consume the MCP Server we created.

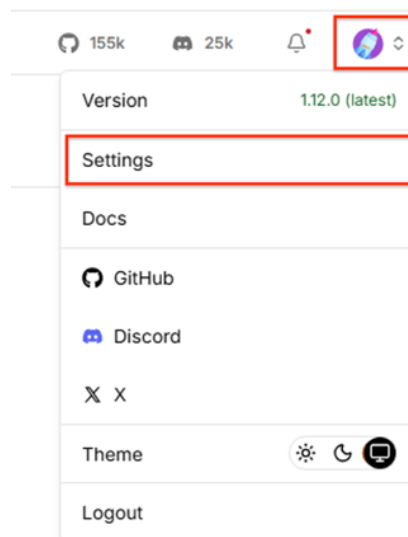
- In your Podman Desktop, you will see that a new container is running.



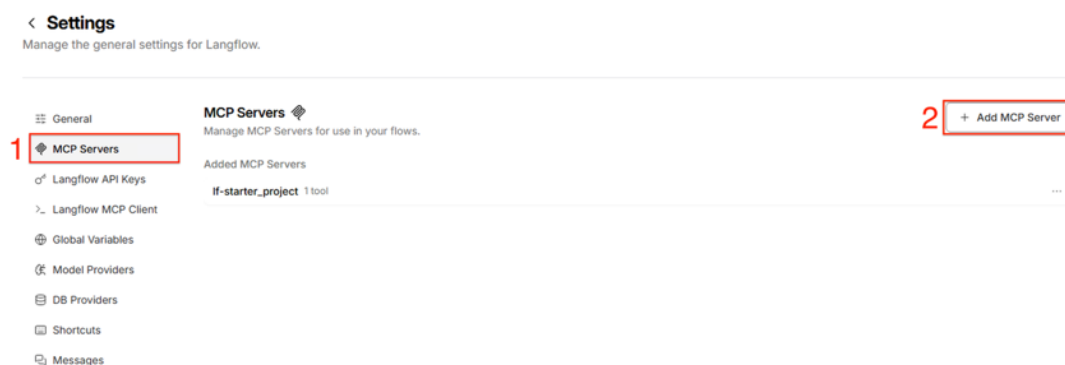
- Langflow is now available at: <http://localhost:7860/>

```
Username: langflow
Password: langflow123
```

- Once you access Langflow, in the upper right corner, access the icon  and settings.



- Access **"MCP Server"** then click on **"+ Add MCP Server"**



- Enter the JSON in the field, and click "Add Server".

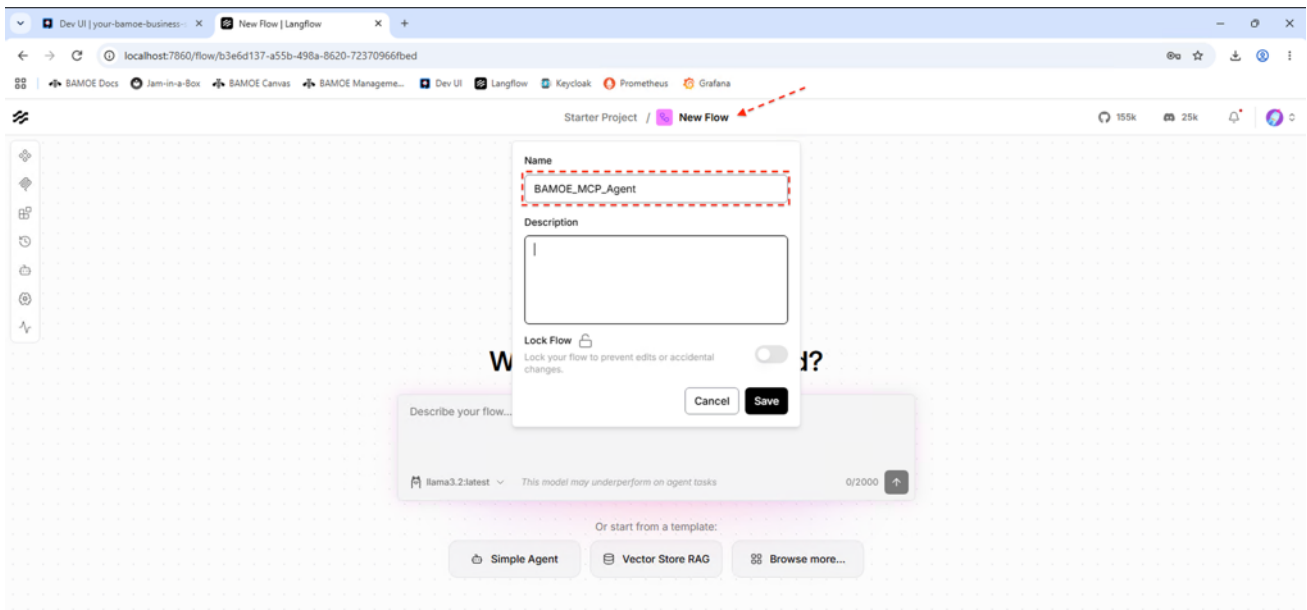
```
{
  "mcpServers": {
    "bamoe-mcp-local": {
      "command": "npx",
      "args": [
        "-y",
        "mcp-remote",
        "http://mcp-server:8080/mcp/sse",
        "--",
        "--allow-http"
      ]
    }
  }
}
```



- Check if your server status shows **"6 tool"**.

bamoe-mcp-local 6 tools

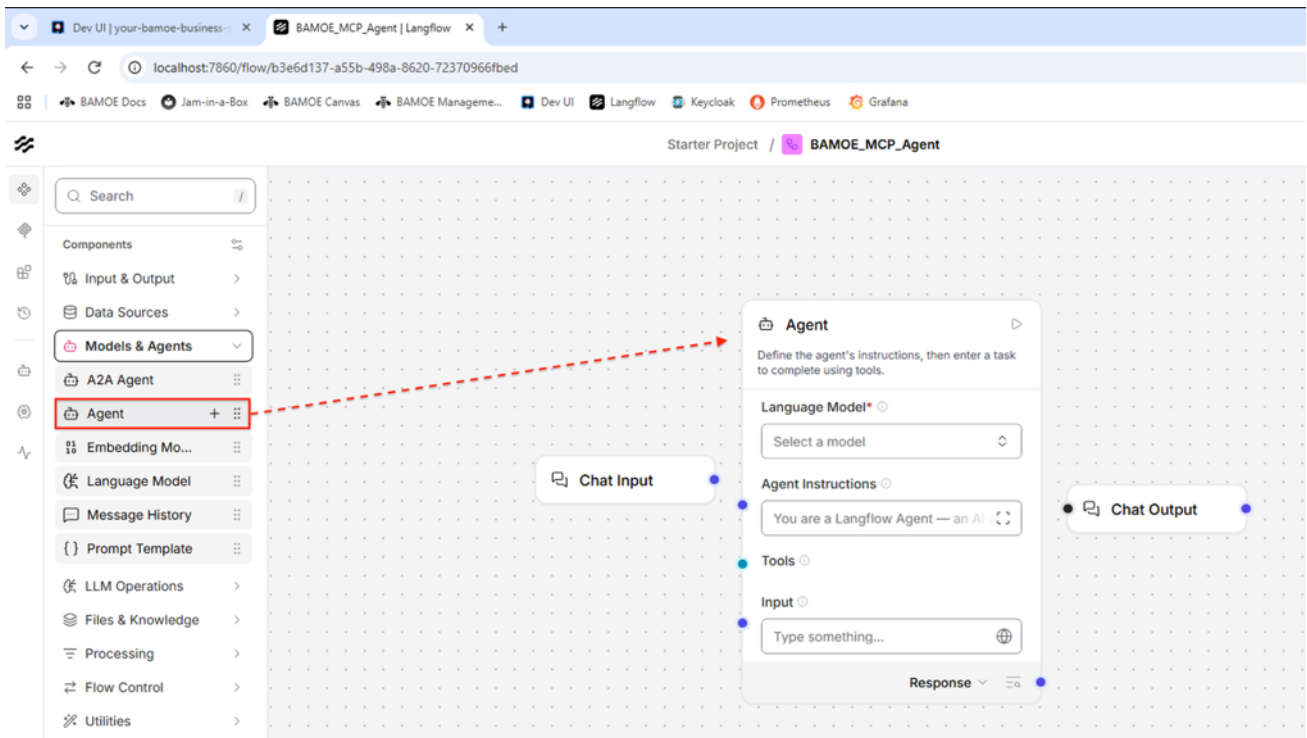
- Click on the **< Settings** to return to the homepage.
- Let's take this opportunity to create a simple agent in Langflow, which we will use in the workflow. On the Langflow homepage, click the **+ Create first flow** button.
- Click the **+ Blank Flow** button.
- Let's rename the flow to **"BAMOE MCP Agent"**



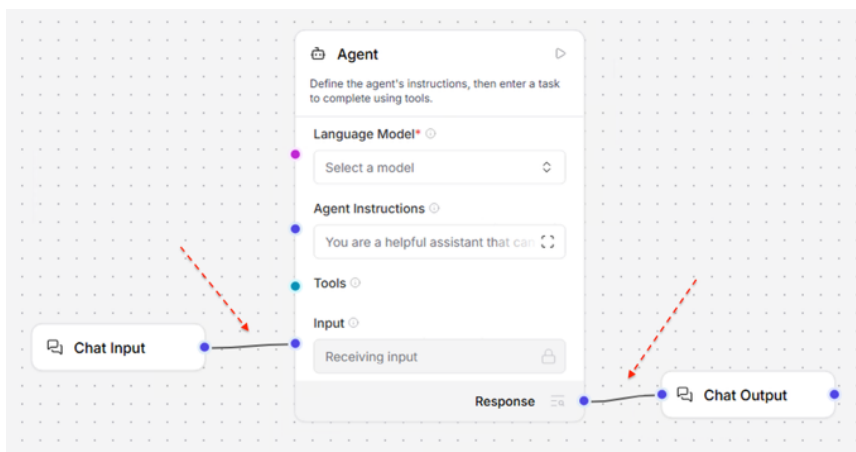
- Now let's create a basic flow for this agent. In **"Input & Output"**, click the + or click and drag a **"Chat Input"** and a **"Chat Output"** to the box.



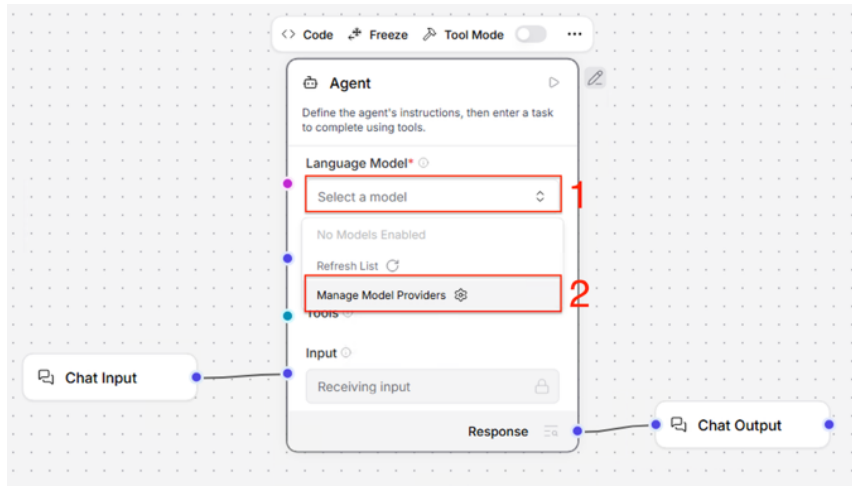
- Now in **"Models & Agents"** add an **"Agent"**.



- Organize the elements in the table and connect them to their respective functions:
 - "Chat Input" connected to "Input Agent"
 - "Response Agent" to "Chat Output"



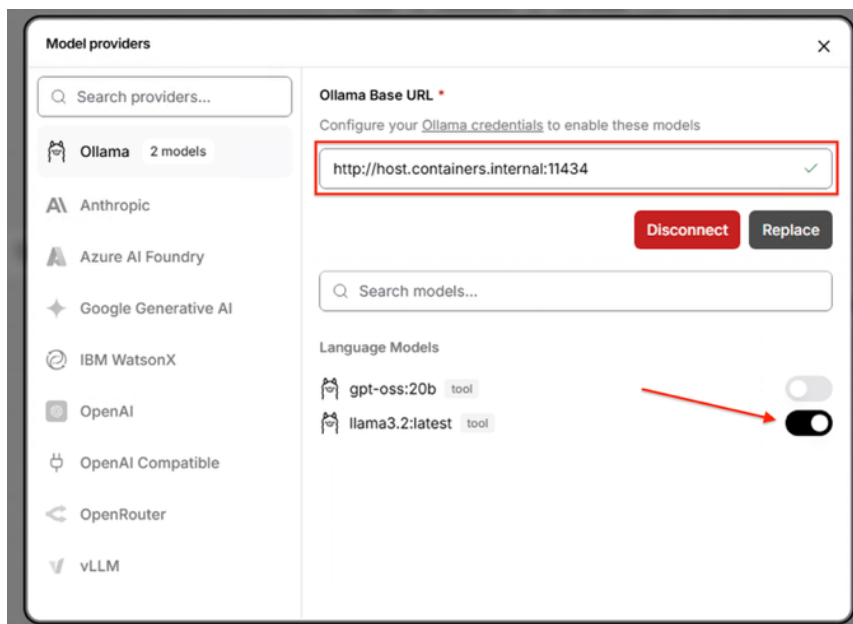
- To configure the agent, click **Select Model > Manage Model Providers**.



- Select Ollama, and configure the directory <http://host.containers.internal:11434>

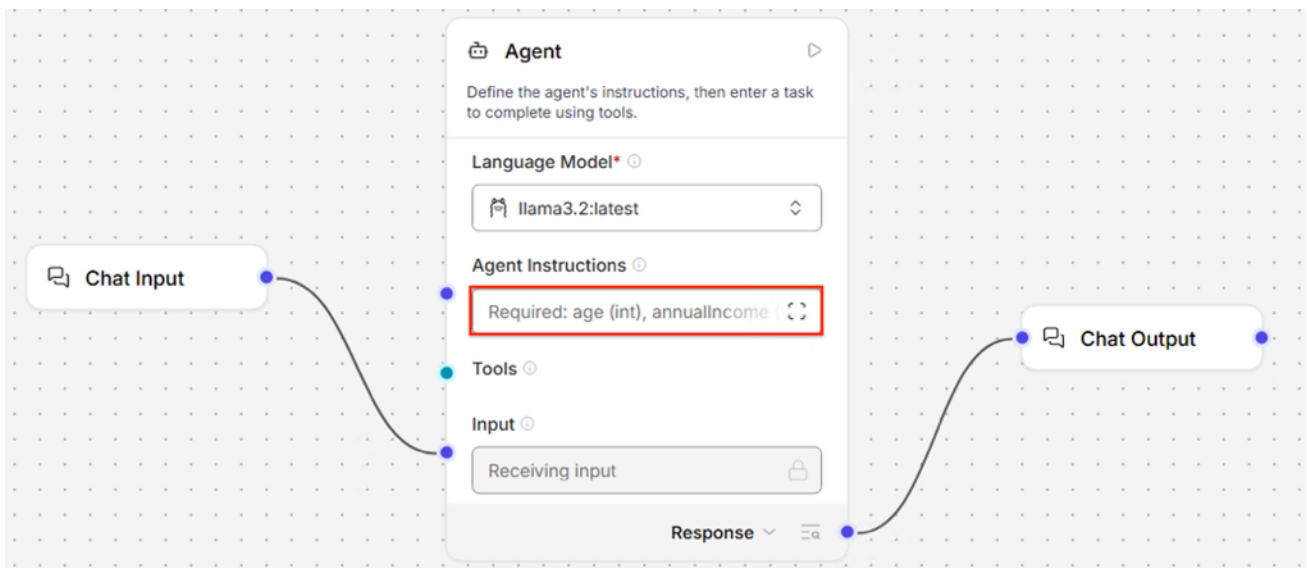
Note: Do not use "localhost:11434", because Langflow is running in a Podman container, and "localhost" doesn't work for communication between the container and the local Windows host.

- Check if the model was displayed, and you can close the window to return to the flow.

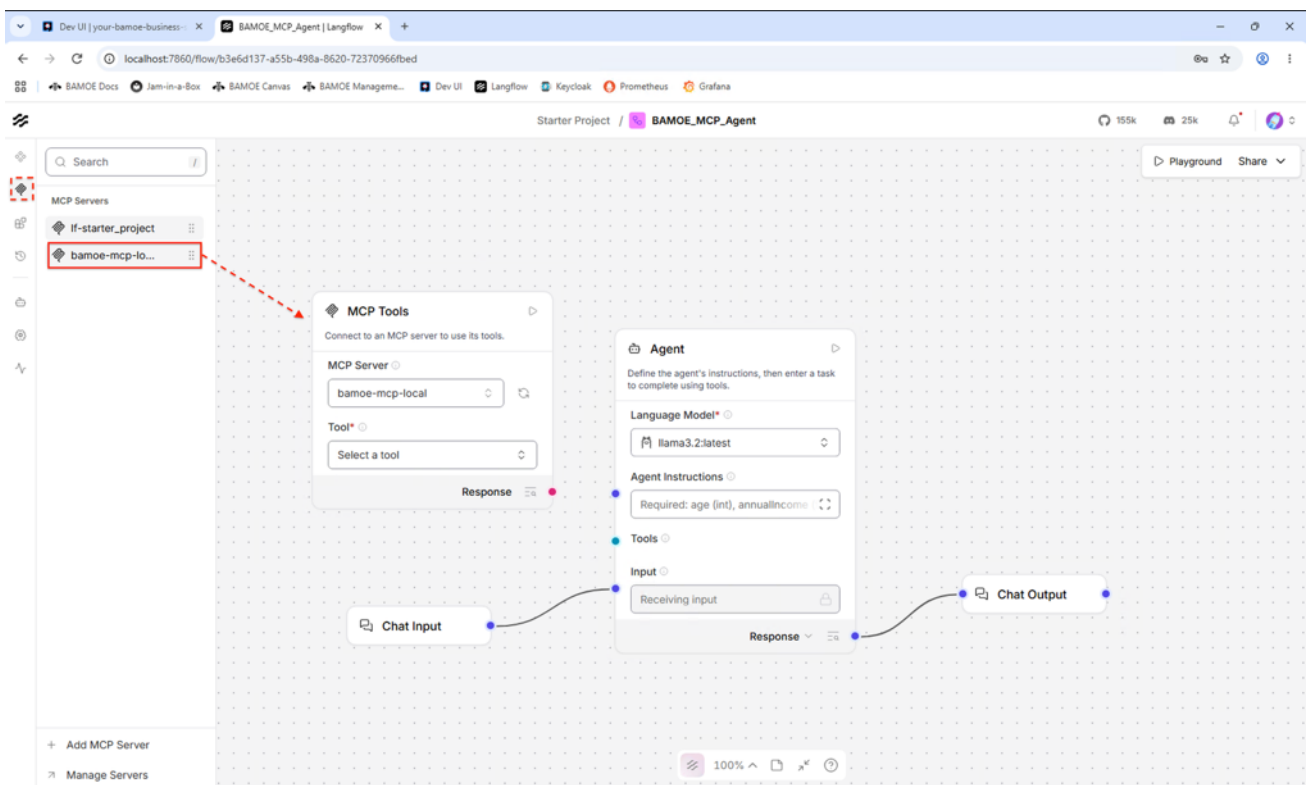


- Add this prompt to the **"Agent Instructions"** field.

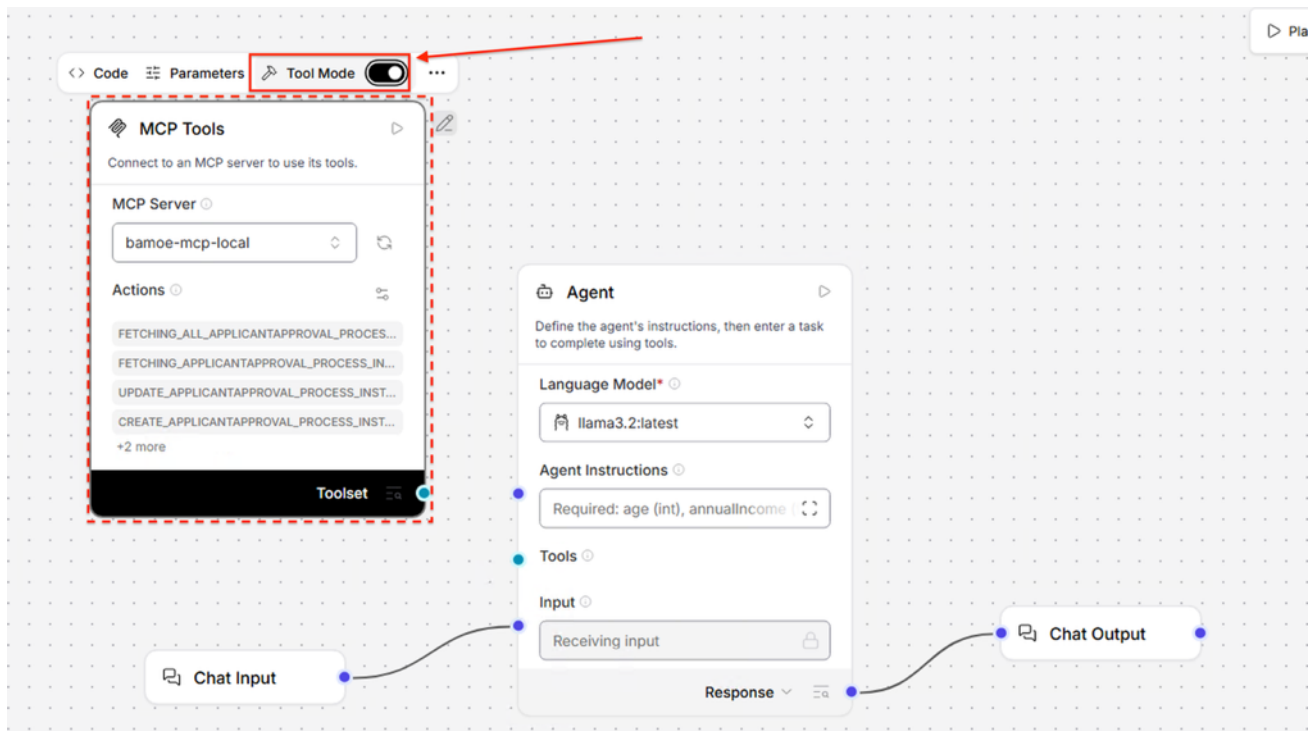
You are a financial agent checking credit card approval; you must obtain the following information before calling the MCP Tool: age (int), annualIncome (float), creditScore (int), student (bool). Highlight a single word that defines the result, then provide context in a short text of no more than two lines.



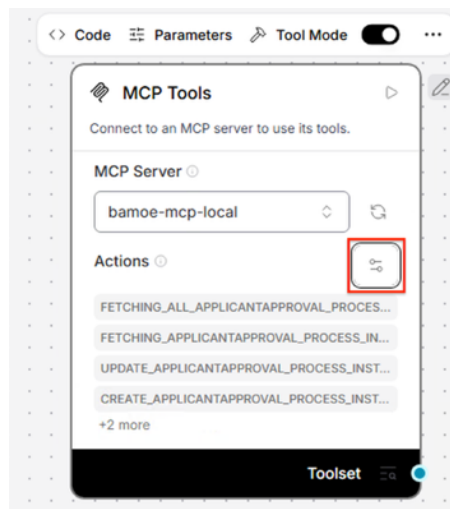
- In the side menu, select MCP Servers and drag the "*bamoe-mcp-server*" we created onto the board.



- Select the MCP Tool component, and in the top toolbar, enable the "***Tool Mode***" setting.



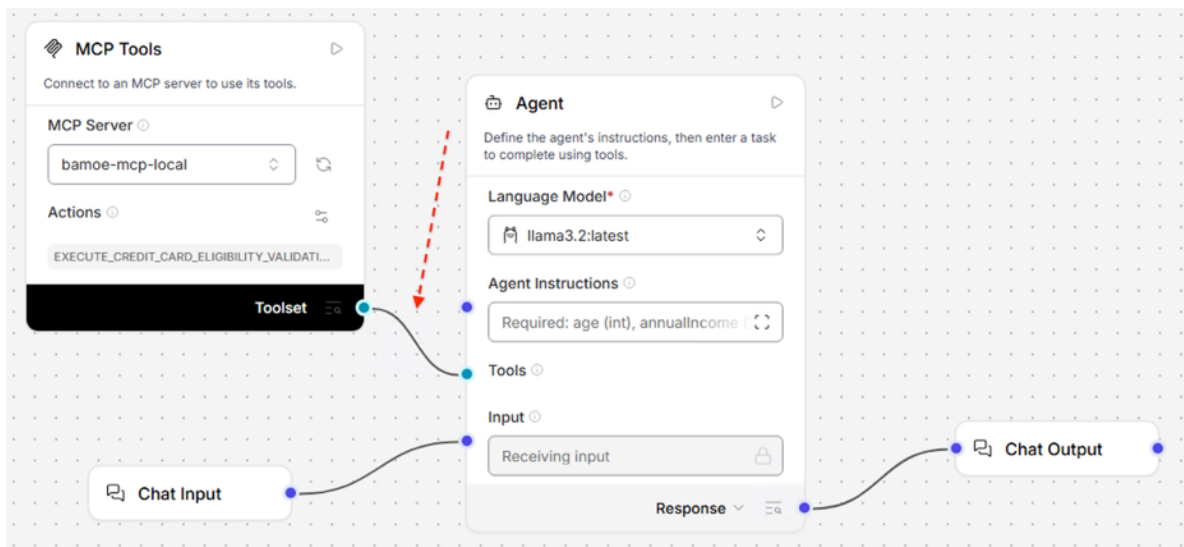
- Now click on **Actions** to specify the tool the agent should use.



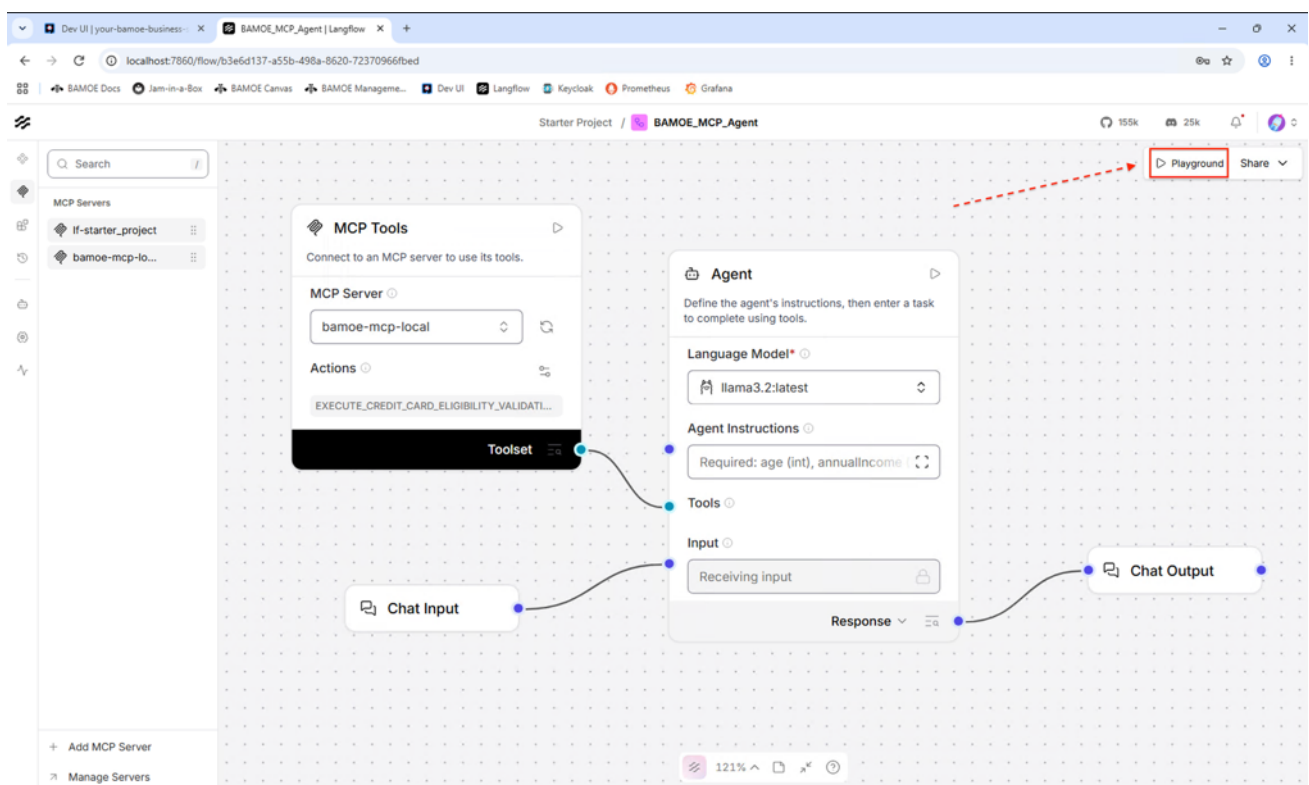
- Select only the "**Execute Credit Card Eligibility Validation Dmn Model**" tool.

MCP Tools			
Search tools...			
Name	Description	Slug	Requires Approval
☐ Fetching All Applicantapproval Process Instance	It fetches all the active "ApplicantApproval" process inst...	FETCHING_ALL_APPLICANTAPPROV...	☐
☐ Fetching Applicantapproval Process Instance	It fetches the active "ApplicantApproval" processes inst...	FETCHING_APPLICANTAPPROVAL_...	☐
☐ Update Applicantapproval Process Instance	It updates an existing "ApplicantApproval" BPMN proce...	UPDATE_APPLICANTAPPROVAL_PR...	☐
☐ Create Applicantapproval Process Instance	It creates a new "ApplicantApproval" BPMN process ins...	CREATE_APPLICANTAPPROVAL_PR...	☐
☒ Execute Credit Card Eligibility Validation Dmn Model	It executes the Credit_Card_Eligibility_Validation DMN ...	EXECUTE_CREDIT_CARD_ELIGIBILIT...	☐
☐ Delete Applicantapproval Process Instance	It deletes an existing "ApplicantApproval" BPMN proces...	DELETE_APPLICANTAPPROVAL_PR...	☐

- Connect the "toolset" to the agent.



- Now that your agent is connected, you can click on "Playground" to interact with the agent.



- See how the agent behaves with this interaction:

John is 35 years old, has an annual income of \$35,000, a credit score of 400, and is not a student. Can he have a credit card?

- In the agent's response, you can expand the view to check the arguments and the result.

The screenshot shows a web application interface with a sidebar on the left containing 'Sessions' and 'Default Session'. The main area displays a chat conversation:

User: John is 35 years old, has an annual income of \$35,000, a credit score of 400, and is not a student. Can he have a credit card?

AI: Finished in 886 | 12.5s

EXECUTE CREDIT CARD ELIGIBILITY VALIDATION DMN 614ms

ARGUMENTS

```
json
{
  "Applicant": {
    "age": 35,
    "annualIncome": 35000,
    "creditScore": 400,
    "student": false
  }
}
```

OUTPUT

Result Metadata

IsEligible result is: manual Eligibility Score Calculation result is: 20

Unfortunately, John does not qualify for a credit card with a credit score of 400. The Eligibility Score Calculation resulted in 20, which is below the threshold for credit card approval.

Send a message...

Congratulations! You have completed this lab.

Stay tuned for new product updates and official IBM documentation. I hope that from now on you will be able to create and explore your own use cases.

6 Consult Documentation and Communities

- [IBM BAMOE Official Documentation](#)
- [IBM Business Automation Community: Open Editions](#)